

Numerical Analysis – Lecture 12

8 The Peano kernel theorem

8.1 The theorem

Our point of departure is the *Taylor formula with an integral remainder term*,

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^k}{k!}f^{(k)}(a) + \frac{1}{k!} \int_a^x (x-\theta)^k f^{(k+1)}(\theta) d\theta, \quad (8.1)$$

which can be verified by integration by parts. Suppose that we are given an approximant (e.g. to a function, a derivative, an integral etc.) whose error vanishes for all $f \in \mathbb{P}_k[x]$. The Taylor formula produces an expression for the error that depends on $f^{(k+1)}$. This is the basis for the *Peano kernel theorem*.

Formally, let $L(f)$ be an error of an approximant. Thus, L maps $C[a, b]$, say, to $\mathbb{R}$. We assume that it is *linear*, i.e. $L(\alpha f + \beta g) = \alpha L(f) + \beta L(g) \forall \alpha, \beta \in \mathbb{R}$, and that $L(f) = 0$ for all $f \in \mathbb{P}_k[x]$. Thus, (8.1) implies

$$L(f) = \frac{1}{k!} L \left\{ \int_a^x (x-\theta)^k f^{(k+1)}(\theta) d\theta \right\}, \quad a \leq x \leq b.$$

To make the range of integration independent of x , we introduce the notation

$$(x-\theta)_+^k := \begin{cases} (x-\theta)^k, & x \geq \theta, \\ 0, & x \leq \theta, \end{cases} \quad \text{whence} \quad L(f) = \frac{1}{k!} L \left\{ \int_a^b (x-\theta)_+^k f^{(k+1)}(\theta) d\theta \right\}.$$

Let $K(\theta) := L[(x-\theta)_+^k]$ for $x \in [a, b]$. [Note: K is independent of f .] Suppose that it is allowed to exchange the order of action of $\int$ and L . Because of the linearity of L , we then have

$$L(f) = \frac{1}{k!} \int_a^b K(\theta) f^{(k+1)}(\theta) d\theta. \quad (8.2)$$

The Peano kernel theorem Let L be a *linear functional* (a linear mapping from a space of functions to $\mathbb{R}$) such that $L(f) = 0$ for all $f \in \mathbb{P}_k[x]$. Provided that $f \in C^{k+1}[a, b]$ and the above exchange of L with the integration sign is valid, the formula (8.2) is true. $\square$

8.2 An example and few useful formulae

Let $L(f) := f'(0) - [-\frac{3}{2}f(0) + 2f(1) - \frac{1}{2}f(2)]$ – this corresponds to approximating

$$f'(0) \approx -\frac{3}{2}f(0) + 2f(1) - \frac{1}{2}f(2).$$

Then $L(f) = 0$ for $f \in \mathbb{P}_2[x]$ (verify by trying $f(x) = 1, x, x^2$ and invoking linearity). Thus, for $f \in C^3[0, 2]$ we have

$$L(f) = \frac{1}{2} \int_0^2 K(\theta) f'''(\theta) d\theta.$$

To evaluate the *Peano kernel* K , we fix θ . Letting $g(x) := (x - \theta)_+^2$, we have

$$\begin{aligned} K(\theta) &= L(g) = g'(0) - \left[-\frac{3}{2}g(0) + 2g(1) - \frac{1}{2}g(2)\right] \\ &= 2(0 - \theta)_+ - \left[-\frac{3}{2}(0 - \theta)_+^2 + 2(1 - \theta)_+^2 - \frac{1}{2}(2 - \theta)_+^2\right] \\ &= \begin{cases} -2\theta + \frac{3}{2}\theta^2 + (2\theta - \frac{3}{2}\theta^2) = 0, & \theta \leq 0, \\ -2(1 - \theta)^2 + \frac{1}{2}(2 - \theta)^2 = 2\theta - \frac{3}{2}\theta^2, & 0 \leq \theta \leq 1, \\ \frac{1}{2}(2 - \theta)^2, & 1 \leq \theta \leq 2, \\ 0, & \theta \geq 2. \end{cases} \end{aligned}$$

[Note: It is obvious that $K(\theta) = 0$ for $\theta \notin [0, 2]$, since then it acts on a quadratic polynomial.]

Back to the general case... Typically, L involves differentiation and integration. Since

$$\frac{d}{dx}(x - \theta)_+^k = k(x - \theta)_+^{k-1}, \quad \int_0^x (t - \theta)_+^k dt = \frac{1}{k+1}[(x - \theta)_+^{k+1} - (a - \theta)_+^{k+1}],$$

the exchange of L with integration is justified in these cases.

Theorem Suppose that K doesn't change sign in (a, b) and that $f \in C^{k+1}[a, b]$. Then

$$L(f) = \frac{1}{k!} \left[\int_a^b K(\theta) d\theta \right] f^{(k+1)}(\xi) \quad \text{for some } \xi \in (a, b).$$

Proof. Let (perversely!) $K \leq 0$. Then

$$L(f) \leq \frac{1}{k!} \int_a^b K(\theta) \min_{x \in [a, b]} f^{(k+1)}(x) d\theta = \frac{1}{k!} \left(\int_a^b K(\theta) d\theta \right) \min_{x \in [a, b]} f^{(k+1)}(x).$$

Likewise $L(f) \geq \frac{1}{k!} \left[\int_a^b K(\theta) d\theta \right] \max_{x \in [a, b]} f^{(k+1)}(x)$, consequently

$$\min_{x \in [a, b]} f^{(k+1)}(x) \leq \frac{L[f]}{\frac{1}{k!} \int_a^b K(\theta) d\theta} \leq \max_{x \in [a, b]} f^{(k+1)}(x)$$

and the required result follows from the mean value theorem. Similar analysis pertains to the case $K \geq 0$. $\square$

Back to our example We have $K \geq 0$ and $\int_0^2 K(\theta) d\theta = \frac{2}{3}$. Consequently $L(f) = \frac{1}{2!} \times \frac{2}{3} f'''(\xi) = \frac{1}{3} f'''(\xi)$ for some $\xi \in (0, 2)$. We deduce in particular that $|L(f)| \leq \frac{1}{3} \|f'''\|_\infty$, where $\|g\|_\infty := \max_{x \in [0, 2]} |g(x)|$ – the ∞ -norm.

Likewise, generalising the definition of the ∞ -norm to an arbitrary interval $[a, b]$, we can easily deduce from

$$\left| \int_a^b f(x)g(x) dx \right| \leq \|g\|_\infty \int_a^b |f(x)| dx,$$

that $|L(f)| \leq \frac{1}{k!} \int_a^b |K(\theta)| d\theta \|f^{(k+1)}\|_\infty$ and that $|L(f)| \leq \frac{1}{k!} \|K\|_\infty \int_a^b |f^{(k+1)}(x)| dx$. This is valid also when K changes sign. Moreover, letting $\|f\|_2 := \left[\int_a^b |f(x)|^2 dx \right]^{1/2}$ – the 2 -norm – the *Cauchy-Schwarz inequality* $\left| \int_a^b f(x)g(x) dx \right| \leq \|f\|_2 \|g\|_2$ implies that $|L(f)| \leq \frac{1}{k!} \|K\|_2 \|f^{(k+1)}\|_2$.